MSG No. 22.46 $F2221'$ [Type II, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2221'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,17,18,19,20]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,7,8,21,22,23,24]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,11,12,25,26,27,28]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[13,14,15,16,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `2221'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[5,6,7,8,21,22,23,24]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,10,11,12,25,26,27,28]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,15,16,29,30,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,6,11,16,17,22,27,32]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,5,12,15,18,21,28,31]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,8,9,14,19,24,25,30]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,7,10,13,20,23,26,29]`

Table 4: Wyckoff site: `4d`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,6,11,16,17,22,27,32]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,5,12,15,18,21,28,31]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,8,9,14,19,24,25,30]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[4,7,10,13,20,23,26,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,17,18]`
2	$[-x,\ 0,\ 0]$	`[3,4,19,20]`
3	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,21,22]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,23,24]`
5	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,25,26]`
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[11,12,27,28]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[13,14,29,30]`

8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[15,16,31,32]

Table 6: Wyckoff site: 8f, site symmetry: .2.1'

No.	position	mapping
1	$[0,\ y,\ 0]$	[1,3,17,19]
2	$[0,\ -y,\ 0]$	[2,4,18,20]
3	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,7,21,23]
4	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[6,8,22,24]
5	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	[9,11,25,27]
6	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[10,12,26,28]
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[13,15,29,31]
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[14,16,30,32]

Table 7: Wyckoff site: 8g, site symmetry: ..21'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,4,17,20]
2	$[0,\ 0,\ -z]$	[2,3,18,19]
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[5,8,21,24]
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[6,7,22,23]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,12,25,28]
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[10,11,26,27]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[13,16,29,32]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[14,15,30,31]

Table 8: Wyckoff site: 8h, site symmetry: ..21'

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,16,17,32]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[2,15,18,31]
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[3,14,19,30]
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[4,13,20,29]
5	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[5,12,21,28]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[6,11,22,27]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[7,10,23,26]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[8,9,24,25]

Table 9: Wyckoff site: 8i, site symmetry: .2.1'

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	[1,11,17,27]
2	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	[2,12,18,28]
3	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	[3,9,19,25]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	[4,10,20,26]
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[5,15,21,31]
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[6,16,22,32]
7	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[7,13,23,29]
8	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[8,14,24,30]

Table 10: Wyckoff site: 8j, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,6,17,22]
2	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[2,5,18,21]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,8,19,24]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,7,20,23]
5	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[9,14,25,30]
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	[10,13,26,29]
7	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[11,16,27,32]
8	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[12,15,28,31]

Table 11: Wyckoff site: 16k, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,17]
2	$[x,\ -y,\ -z]$	[2,18]
3	$[-x,\ y,\ -z]$	[3,19]
4	$[-x,\ -y,\ z]$	[4,20]
5	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[5,21]
6	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[6,22]
7	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7,23]
8	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[8,24]
9	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[9,25]
10	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[10,26]
11	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[11,27]
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[12,28]
13	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[13,29]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[14,30]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[15,31]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[16,32]